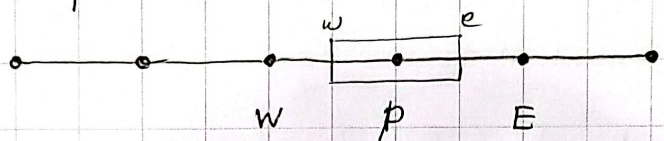
Generalized FVM Discretization.

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$$\frac{d}{dx}\left(k \cdot \frac{dT}{dx}\right) = 0$$

Consider finite volume mesh:



Integrate over finite volume

$$\int \frac{d}{dx}\left(k \cdot \frac{dT}{dx}\right) dV = 0 .$$

|| Gauss divergence thm: converts a volume integral into a surface integral.

$$\int \vec{\nabla} \cdot \left(k \frac{dT}{dx}\right) dA = 0$$

more general gauss thm:

$$\int div\left(\Gamma \cdot \overrightarrow{grad}\,\phi\right) d\bar{V} = \int \hat{n} \cdot \left(\Gamma \overrightarrow{grad}\,\phi\right) dA .$$

$$\int div\left(\vec{F}\right) dV = \int \vec{F} \cdot \hat{n}\, dA .$$

$\hat{n}$ → outward normal vector.

face not involved in $\pm$ -1).

face involved ← | → face involved

w - - - - - e

↓ face not involved in 1-D.

Only contribution from the east face and west face.
Bottom and Top face not involved.

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$$kA \cdot \left.\frac{dT}{dx}\right|_e - kA \left.\frac{dT}{dx}\right|_w = 0 .$$

k, A can be variables.

$$k_e \cdot A_e \cdot \underbrace{\left.\frac{dT}{dx}\right|_e}_{\text{using Central differencing.}} - k_w A_w \underbrace{\left.\frac{dT}{dx}\right|_w}_{\text{using Central differencing.}} = 0 .$$

$$k_e A_e \left(\frac{T_E - T_P}{h_{PE}}\right) - k_w A_w \left(\frac{T_P - T_W}{h_{WP}}\right) = 0$$

Rearranging.

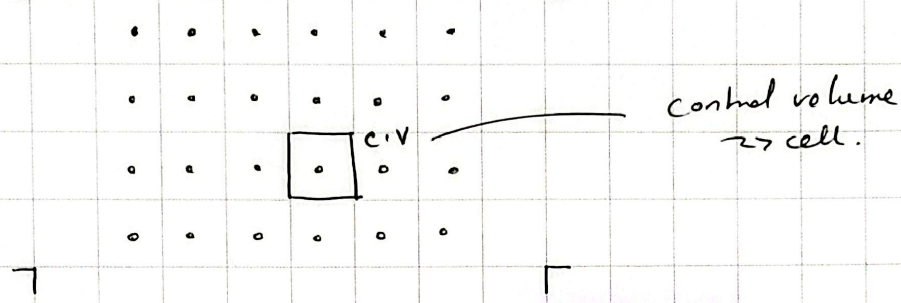
$$a_P T_P = a_W T_W + a_E T_E .$$

$$a_W = \frac{k_w A_w}{h_{WP}} \qquad a_E = \frac{k_e A_e}{h_{PE}}$$

$$a_P = a_W + a_E = \frac{k_w A_w}{h_{WP}} + \frac{k_e A_e}{h_{PE}} .$$

2D - FVM. General

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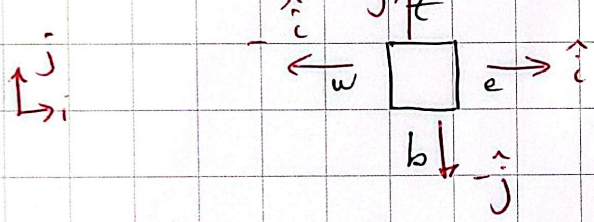
Consider Pure diffusion.

$$\text{e.g. } \nabla\cdot(\Gamma \text{grad} T) = 0.$$

Integrate over control volume.

$$\int \nabla\cdot(\Gamma \text{grad} T)\, dV = \int \hat{n}\cdot(\Gamma \text{grad} T)\, dA.$$

→ normal vector perpendicular to surface (out)



4 faces involved in transport of T.

$$\int \Gamma \frac{dT}{dx}\, dA_e + \int -\Gamma \frac{dT}{dx}\, dA_w + \int \Gamma \frac{dT}{dy}\, dA_t + \int -\Gamma \frac{dT}{dy}\, dA_b = 0$$

$\hat{n}_e = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$; $\text{grad} T = \begin{pmatrix} \partial_x T \\ \partial_y T \end{pmatrix}$ ⇒ $\partial_x T$

$$\Gamma A \cdot \frac{dT}{dx}\Big|_e - \Gamma A \frac{dT}{dx}\Big|_w + \Gamma A \frac{dT}{dy}\Big|_t - \Gamma A \frac{dT}{dy}\Big|_b = 0$$

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